

EXAMPLE (CONTINUED)

Grading oral exams. (Grades are given immediately) like uhk online streaming algo for y'all CS needs
The prof sees n students one-by-one, and wants to give one A^+ to the best student

Strategy: Give an A^+ to the first student $\Rightarrow P(\text{success}) = P(\text{first student is the best}) = \frac{1}{n}$ (not good)

Better strategy: See the first k students, and give no A^+ s. Give an A^+ to a student if they are better than everyone before them

Define $S = \begin{cases} 1, & \text{we give } A^+ \text{ to the best student} \\ 0, & \text{o/w} \end{cases} \Rightarrow P(\text{success}) = E[S]$

Let X = position of best student

By prop, $P(\text{success}) = E[S] = E[E[S|X]] = \sum_{i=1}^n E[S|X=i] P(X=i) = \sum_{i=1}^n E[S|X=i] (\frac{1}{n})$ (continue next lecture)
 $= \frac{1}{n} \sum_{i=1}^n P(\text{best student gets } A^+ | \text{they are } i^{\text{th}})$
 $= \frac{1}{n} \sum_{i=1}^k P(\text{best student gets } A^+ | \text{they are } i^{\text{th}}) + \sum_{i=k+1}^n P(\text{best student gets } A^+ | \text{they are } i^{\text{th}})$ $\leftarrow \because \text{first } k \text{ have no } A^+$

This problem becomes: $\star$: best among bracket, $\star$: best overall

$(\underbrace{\quad \quad \quad}_k \star \mid \underbrace{\quad \quad \quad}_{i^{\text{th}}}) \star$ (can be ignored since def A^+ is given by now)

In the first $i-1$ students, where is the best among them ($\star$)? Is it in $k+1$ to $i-1$?

Thus, best student gets $A^+ | i^{\text{th}} \text{ pos} \Leftrightarrow \text{best of first } i-1 \text{ students is in the first } k \text{ pos}$
 $\Rightarrow P(A^+ | i^{\text{th}}) = \frac{k}{i-1}$

$\therefore P(\text{success}) = \frac{1}{n} \sum_{i=k+1}^n \frac{k}{i-1} = \frac{k}{n} \sum_{i=k}^{n-1} \frac{1}{i} = \frac{k}{n} (\sum_{i=1}^{n-1} \frac{1}{i} - \sum_{i=1}^{k-1} \frac{1}{i}) \sim \frac{k}{n} (\log n - \log k) = \frac{k}{n} \log \frac{n}{k}$ $\leftarrow \text{harmonic series}$

How can prof optimize choice of k w/ this strategy?

$$\frac{dP(\text{success})}{dk} = \frac{1}{n} \log \frac{n}{k} + \frac{k}{n} \left(\frac{1}{n/k} \right) \left(-\frac{n}{k^2} \right) = \frac{1}{n} \log \frac{n}{k} - \frac{1}{n}$$

$$\frac{dP(\text{success})}{dk} = 0 \Rightarrow \log \frac{n}{k} = 1 \Rightarrow k = \frac{n}{e}$$

MOMENT GENERATING FUNCTIONS**DEFINITION**

Given a rand var X , its moment generating function is $M_X(t) := E[e^{tx}] \quad \forall t \in \mathbb{R}$

PROPOSITION

Given a rand var X with its moment generating function M_X , and some $n \in \mathbb{N}$, the n^{th} moment of X is $E[X^n] = \frac{d^n}{dt^n} M_X(0)$
 for all d ! we care about, we can exchange the expectation and differentiation $(\frac{d}{dt} \int x P(X=x) dx = \int x \frac{d}{dt} P(X=x) dx)$

Proof

$$\frac{d^n}{dt^n} M_X(0) = \frac{d^n}{dt^n} E[e^{tx}] \Big|_{t=0} = E \left(\frac{d^n}{dt^n} (e^{tx}) \right) \Big|_{t=0} = E[X^n e^{tx}] \Big|_{t=0} = E[X^n e^{t \cdot 0}] = E[X^n] \quad \square$$

FACT

If two rand var have the same moment generating function, then they have the same distribution

EXAMPLE

$X \sim \text{Bin}(n, p)$

Then, by def, $M_X(t) = E[e^{tx}] = \sum_{k=0}^n e^{tk} P(X=k) = \sum_{k=0}^n e^{tk} \binom{n}{k} p^k (1-p)^{n-k} = \sum_{k=0}^n \binom{n}{k} (pe^t)^k (1-p)^{n-k}$

By binomial thm, $= (1-pe^t)^n$

Note, $M_X(t) = (1 - p + pe^t)^n$

$$M_X'(t) = n(1 - p + pe^t)^{n-1} pe^t \Rightarrow \underline{E[X] = M_X'(0) = n(1 - p + p)^{n-1} p = np}$$

Note, $M_X''(t) = n(n-1)(1 - p + pe^t)^{n-2} p^2 e^{2t} + n(1 - p + pe^t)^{n-1} pe^t \Rightarrow \underline{E[X^2] = M_X''(0) = n(n-1)p^2 + np}$

$$\therefore \text{Var}(X) = E[X^2] - E[X]^2 = n(n-1)p^2 + np - n^2p^2 = np(1-p)$$

EXAMPLE

$X \sim \text{Poi}(\lambda)$

Then, $M_X(t) = E[e^{tx}] = \sum_{k=0}^{\infty} e^{tk} \frac{e^{-\lambda} \lambda^k}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!} = e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t - 1)}$

$$M_X'(t) = e^{\lambda(e^t - 1)} \cdot \lambda e^t \Rightarrow E[X] = M_X'(0) = \lambda$$

$$M_X''(t) = \lambda^2 e^{2t} e^{\lambda(e^t - 1)} + \lambda e^t e^{\lambda(e^t - 1)} \Rightarrow \underline{E[X^2] = M_X''(0) = \lambda^2 + \lambda}$$

$$\therefore \text{Var}(X) = E[X^2] - E[X]^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

TL;DR: MOMENT GENERATING FUNCTIONS GIVE US AN EASIER WAY TO COMPUTE $E[X^n]$

PROPOSITION

If X, Y indep, then $M_{X+Y}(t) = M_X(t) M_Y(t)$

Proof
By def, $M_{X+Y}(t) = E[e^{t(X+Y)}] = \underline{E[e^{tX} e^{tY}]} = \underline{E[e^{tX}] E[e^{tY}]} = M_X(t) M_Y(t) \quad \square$ (indep)

COROLLARY

If $X \sim \text{Poi}(\lambda_1), Y \sim \text{Poi}(\lambda_2)$ are indep, then $X_1 + X_2 \sim \text{Poi}(\lambda_1 + \lambda_2)$

Proof
 $M_X(t) = e^{\lambda_1(e^t - 1)}$ and $M_Y(t) = e^{\lambda_2(e^t - 1)}$
Since X, Y indep, hence $M_{X+Y}(t) = M_X(t) M_Y(t) = e^{\lambda_1(e^t - 1)} e^{\lambda_2(e^t - 1)} = e^{(\lambda_1 + \lambda_2)(e^t - 1)}$, which is the moment generating function of $\text{Poi}(\lambda_1 + \lambda_2)$

By uniqueness of m.g.f. to distributions, thus $X+Y \sim \text{Poi}(\lambda_1 + \lambda_2) \quad \square$

EXAMPLE

$X \sim \text{Exp}(\lambda)$

Then, $M_X(t) = E[e^{tx}] = \int_0^{\infty} e^{tx} f_X(x) dx = \int_0^{\infty} e^{tx} \lambda e^{-\lambda x} dx = \lambda \left[-\frac{e^{-(\lambda - t)x}}{\lambda - t} \right]_0^{\infty} = \frac{\lambda}{\lambda - t}$ (t < λ)

$$\hookrightarrow M_X'(t) = \frac{\lambda}{(\lambda - t)^2} \Rightarrow E[X] = M_X'(0) = \frac{1}{\lambda}$$

$$\hookrightarrow M_X''(t) = \frac{2\lambda}{(\lambda - t)^3} \Rightarrow \underline{E[X^2] = M_X''(0) = \frac{2}{\lambda^2}} \Rightarrow \text{Var}(X) = E[X^2] - E[X]^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

EXAMPLE

$Z \sim N(0, 1)$

Then, $M_Z(t) = E[e^{tz}]] = \int_{-\infty}^{\infty} e^{tz} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz = \frac{1}{\sqrt{2\pi}} e^{\frac{t^2}{2}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z - t)^2} dz$

Let $x = z - t, \Rightarrow e^{\frac{1}{2}t^2} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}x^2} dx = e^{\frac{1}{2}t^2}$
(density for $N(0, 1)$)

$$\therefore M_Z'(t) = t e^{\frac{1}{2}t^2} \Rightarrow E[Z] = M_Z'(0) = 0$$

$$M_Z''(t) = e^{\frac{1}{2}t^2} + t^2 e^{\frac{1}{2}t^2} \Rightarrow E[Z^2] = M_Z''(0) = 1 \Rightarrow \text{Var}(Z) = E[Z^2] - E[Z]^2 = 1$$

EXAMPLE

$X \sim N(\mu, \sigma^2)$

Then, $\frac{X - \mu}{\sigma} = Z \sim N(0, 1) \Rightarrow X = \mu + \sigma Z$

$$\therefore M_X(t) = E[e^{tx}] = E[e^{t(\mu + \sigma Z)}] = e^{t\mu} E[e^{t\sigma Z}] = e^{t\mu} M_Z(t\sigma) = e^{t\mu} e^{\frac{1}{2}(t\sigma)^2} = e^{t\mu + \frac{1}{2}\sigma^2 t^2}$$

EXAMPLE / COROLLARY

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Let $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$ be indep

By indep, $M_{X+Y}(t) = M_X(t) M_Y(t) = e^{t\mu_1 + \frac{1}{2}t^2\sigma_1^2} e^{t\mu_2 + \frac{1}{2}t^2\sigma_2^2} = e^{t(\mu_1 + \mu_2) + \frac{1}{2}t^2(\sigma_1^2 + \sigma_2^2)}$, which is the m.g.f. of $N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$ $\square$

EXAMPLE

Let $X_1, X_2, X_3, \dots$ be indep identically distributed rand vars

Let N be an indep rand var taking integer nonneg vals

Let $X = \sum_{i=1}^N X_i$, then $E[X] = E[N]E[X_1]$

Proof

By def, $E[e^{tX}] = E[E[e^{t \sum_{i=1}^N X_i} | N=n]] = E_{N=n}[E[e^{t \sum_{i=1}^N X_i}]] = E_{N=n}[\overset{\text{indep}}{E[e^{tX_1}]^N}] = E_{N=n}[\overset{X_i: \text{ i.i.d.}}{M_{X_1}(t)^N}] = E[M_{X_1}(t)^N]$

Thus, $M_X'(t) = E[\frac{d}{dt} M_X(t)^N] = E[N M_{X_1}'(t) M_{X_1}(t)^{N-1}]$

$\therefore E[X] = M_X'(0) = E[N \cdot M_{X_1}'(0) M_{X_1}(0)^{N-1}]$

Notice $M_{X_1}(0) = E[e^{0X_1}] = 1$

$\therefore E[X] = E[N \cdot 1 \cdot M_X'(0)] = E[N E[X_1]] = E[N] E[X_1] \quad \square$

LIMIT THEOREMS

If $X_1, X_2, X_3, \dots$ are indep and identically distributed, we can define the sample mean $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. For convenience, denote $\mu := E[X_1]$

QUESTION

What can we say about the distribution of $\bar{X}_n$?

RECALL

By lin of exp, $E[\bar{X}_n] = \mu$ and by indep, $\text{Var}(\bar{X}_n) = \frac{1}{n} \text{Var}(X_1)$

THREE TYPES OF RESULTS

① Weak law of large numbers: $\forall \epsilon > 0, \lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| \geq \epsilon) = 0$

② Strong law of large numbers: $P(\lim_{n \rightarrow \infty} \bar{X}_n = \mu) = 1$

③ Central Limit Theorem: As $n \rightarrow \infty$, $\frac{\bar{X}_n - \mu \sqrt{n}}{\sigma} \approx N(0, 1)$